

Name
Date
Course

The Law of Cosines (Homework)

Find the missing side lengths and angle measures of each triangle. In other words, solve each triangle.

① $A = 28^\circ$, $b = 6 \text{ ft}$, $c = 5 \text{ ft}$

② $c = 10 \text{ cm}$, $a = 4 \text{ cm}$, $B = 118^\circ$

$$\textcircled{3} \ b = 1.1 \text{ in}, C = 80^\circ, a = 2.7 \text{ in}$$

$$\textcircled{4} \ a = 6 \text{ m}, b = 3 \text{ m}, c = 8 \text{ m}$$

⑤ $a = 5.2 \text{ km}$, $c = 3.8 \text{ km}$, $b = 4.4 \text{ km}$

⑥ $b = 9 \text{ mi}$, $c = 7 \text{ mi}$, $a = 4 \text{ mi}$

Homework Answers

① $a \approx 2.8325 \text{ ft}$
 $C \approx 55.967^\circ$
 $B \approx 96.03^\circ$

② $b \approx 12.3918 \text{ cm}$
 $A \approx 16.5594^\circ$
 $C \approx 45.44^\circ$

③ $c \approx 2.7329 \text{ in}$
 $B \approx 23.3529^\circ$
 $A \approx 76.647^\circ$

④ $C \approx 121.855^\circ$
 $B \approx 18.57^\circ$
 $A \approx 39.57^\circ$

⑤ $A \approx 78.3371^\circ$
 $C \approx 45.699^\circ$
 $B \approx 55.9638^\circ$

⑥ $B \approx 106.6^\circ$
 $A \approx 25.209^\circ$
 $C \approx 48.19^\circ$